

# Sample RAG System & AI Guide

## 1. Retrieval-Augmented Generation Overview

RAG enables Large Language Models to answer questions grounded in PDF documents.

## 2. Document Chunking Parameters

Documents are loaded via PyPDFLoader and split with RecursiveCharacterTextSplitter.

Chunk Size is 500 characters and Chunk Overlap is 50 characters.

## 3. Hugging Face Embeddings & ChromaDB

Text chunks are embedded using all-MiniLM-L6-v2 into 384-dimensional vectors.

ChromaDB persists vector data locally in ./chroma\_db directory.

## 4. Retrieval Top K and Google Gemini Answer

Similarity search retrieves Top K = 3 matching context chunks for Google Gemini LLM.